

Quiz 2

● Graded

Student

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Total Points

10 / 10 pts

Question 1

Transformer

2 / 2 pts

✓ + 2 pts Incorrect

Question 2

Seq2Seq

2 / 2 pts

✓ + 2 pts Correct

Question 3

(no title)

2 / 2 pts

✓ + 2 pts Correct

Question 4

(no title)

2 / 2 pts

✓ + 2 pts Correct

Question 5

(no title)

2 / 2 pts

✓ + 2 pts Correct

Q1 Transformer

2 Points

Which one(s) of the following are TRUE about Transformer models?

☒ In the attention mechanism, the first word will “query” each other word and based on their “value”s to decide its attention on the words.

☒ Attention mechanism allows for focusing on the words with high attention weights, and pay less attention to irrelevant words with low attention weights.

☐ Self-attention will always put a higher attention weight on itself.

☐ Transformers process input tokens sequentially, one at a time.

Q2 Seq2Seq

2 Points

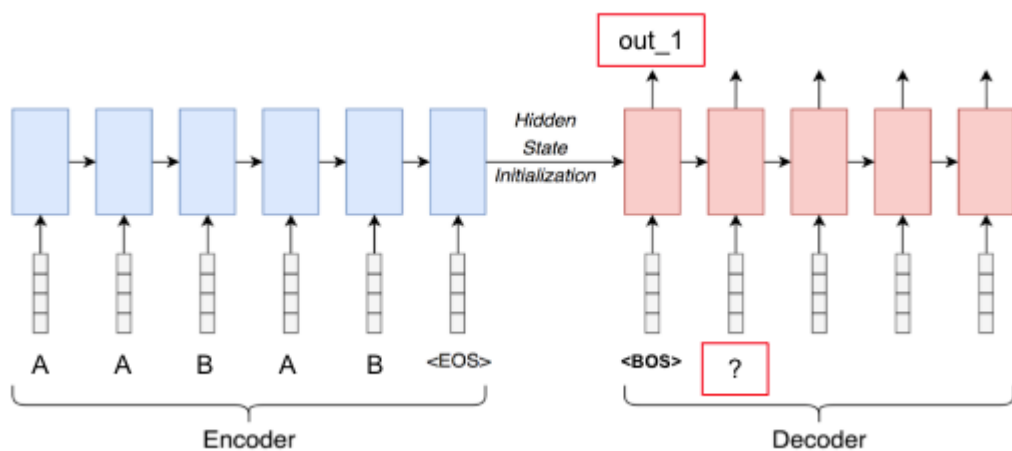
We are using an encoder-decoder RNN at inference (decoding) time. Consider a vocabulary consisting of four tokens: {"A", "B", "", ""}.

At the first decoding step, the decoder receives as input and produces the following probability distribution (out_1):

{"A": 0.7, "B": 0.05, "": 0.15, "": 0.1}

Assume we use greedy decoding.

What is the second input to the decoder (i.e., the token fed at the next time step)?



- ☒ "A"
- ☐ "B"
- ☐ "<BOS>"
- ☐ "<EOS>"

Q3

2 Points

What is the range of the final attention weights in attention mechanisms?

- ☒ 0 to 1
- ☐ -1 to 1
- ☐ 0 to infinity
- ☐ -infinity to infinity

Q4

2 Points

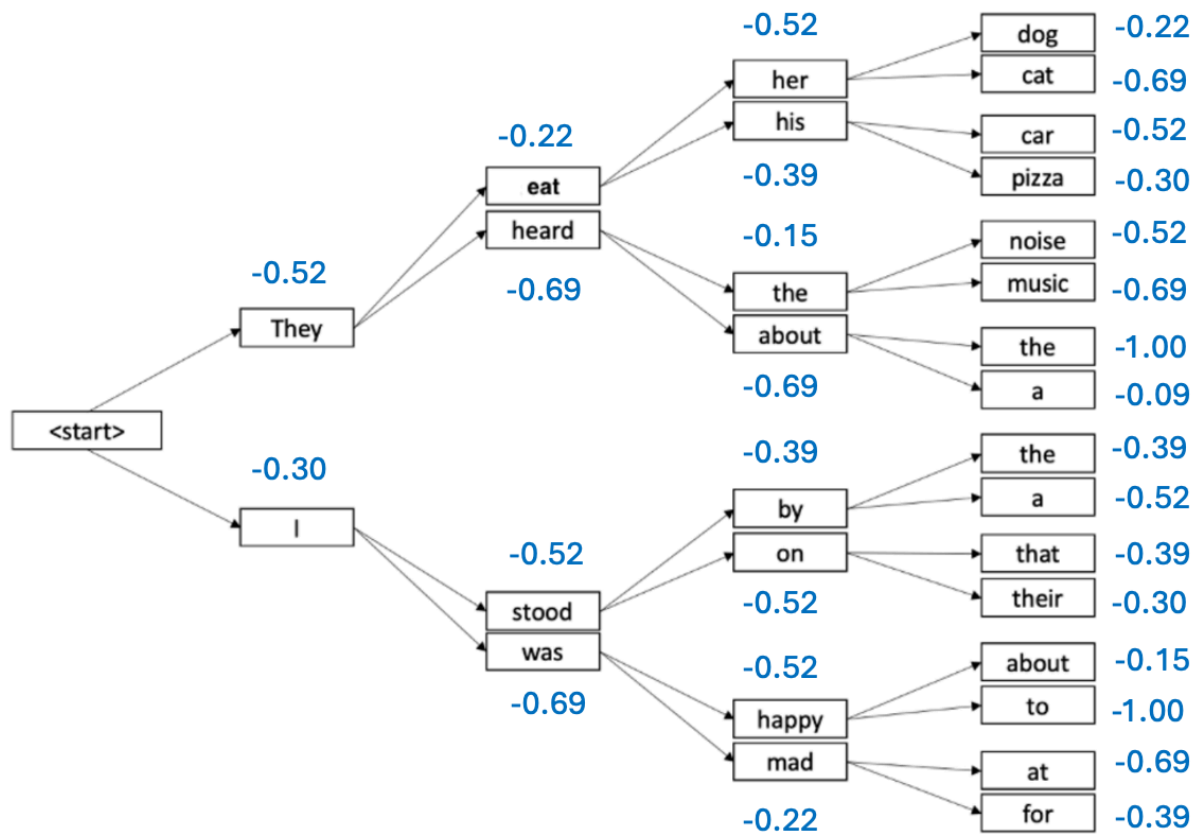
During self-attention computation, if the input sequence length is L and the dimensionality of each input feature is D , what is the space needed for storing the final attention weight matrix?

- ☐ $O(L)$
- ☒ $O(L^2)$
- ☐ $O(D)$
- ☐ $O(LD)$

Q5

2 Points

Given $P(< start >) = 1$, which of the following sequences will be generated if we use beam search with $k=2$?



- ☐ I eat the a
- ☒ They eat his pizza
- ☐ They heard the noise
- ☐ I stood by the